Thanh Dang

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EDUCATION

California State Polytechnic University, Pomona May 2026
Bachelor of Science, Computer Science **GPA:** 3.73 / 4.00

- **Honors:** Magna Cum Laude, STEM Advantage Scholar

WORK EXPERIENCE

CPP Enterprises Pomona, CA
AI/ML Researcher May 2025 – October 2026

- Spearheaded multimodal AI pipeline processing 50+ hours of powered mobility device footage, using WhisperX to transcribe audio into speaker-labeled segments and Google Gemma LLM to generate driving task annotations
- Developed transformer-based multi-task learning framework with PyTorch to analyze upper-body movement asymmetry in stroke survivors during powered mobility; published at IEEE Healthcom Conference 2026
- Model achieved 89% rank accuracy for stroke-healthy control separation and 93% AUC for affected-side prediction

Full Stack Developer January 2026 – May 2026

- Developed clinical web app for NSF-funded mobility research; deployed via Docker Compose on Jetstream2
- Built GPU-accelerated video worker using Ray and Kafka, streaming real-time processing status via SSE and Redis pub/sub; designed PostgreSQL schema and implemented NestJS REST APIs for multi-tenant workflows
- Led development team collaboration with kinesiology researchers to define project scope and deliverables

Cal Poly Pomona STARS Program Pomona, CA
Undergraduate Research Assistant October 2024 – May 2025

- Built multithreaded Python desktop app for data management on HPC cluster via SFTP, enabling directory browsing, upload, download, and deletion; presented at Cal Poly Pomona RSCA conference to 20+ attendees
- Developed pipeline combining YOLO11 pose estimation with IoU person tracking for sway displacement analysis
- Aligned pose, motion, and label data via video reference, applying temporal filtering to reduce dataset size by 15%

theCoderSchool Cerritos, CA
Code Coach January 2024 – February 2025

- Led coding classes for 30+ elementary students, teaching programming through game development in Scratch
- Coached 5 students one-on-one in web development, teaching HTML, CSS, and JavaScript fundamentals
- Supported students through independent project work, providing hands-on debugging help and technical guidance

CPP Software Engineer Association Pomona, CA
Frontend Developer August 2023 - May 2024

- Developed mobile platform to connect school organizations with members using React Native and Express.js
- Utilized CSS and React Navigation to implement responsive user interface and smooth screen transitions
- Collaborated in Agile team of 10+ developers, pair programming on assigned tasks across 2-week sprints

PROJECTS

TrailTalk
Backend Developer September 2025 – December 2025

- Built social web app with TypeScript, React, and Firebase for park enthusiasts, featuring 63 U.S. national parks
- Designed Firestore data model for users, park ratings, posts, comments, and chat with nested subcollections
- Implemented private messaging between users, with in-app notifications, using Firestore real-time listeners

Carbon Closet Cal Poly Pomona Hackathon/Broncohacks
Full Stack Developer April 2025

- Built e-commerce platform for secondhand clothing using Next.js and Firebase; won 3rd place among 60 teams
- Developed user interface with Tailwind CSS for clothing listings, search, token shop, and gacha bundle purchases
- Containerized application with Docker and deployed to AWS EC2, enabling scalable, cloud-based hosting

Recyclable or Not Bot UC Riverside Hackathon/Rosehack
Backend Developer January 2025

- Developed computer vision web app to detect recyclability; awarded best sustainability track among 50+ teams
- Implemented object detection and inference using OpenCV frame capture and YOLOv8 deep learning model
- Integrated Flask API with threaded detection pipeline for non-blocking request handling during live inference

SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
- **Frameworks:** React, React Native, Next.js, NestJS, Flask, PyTorch, Ray
- **Tools & Technologies:** Firebase, AWS, PostgreSQL, MySQL, Redis, Docker, Kafka, Node.js, Git